

Solution	Marks	Remarks
1. (a) $Pt = ml$ $l = \frac{150 \times (5 \times 60)}{0.016}$ $= 2812500 \text{ J kg}^{-1} \approx 2810 \text{ kJ kg}^{-1}$	1M 1A 2	
(b) $C(100 - 22) = m_w c_w (22 - 20)$ $C = \frac{0.100 \times 4200(22 - 20)}{(100 - 22)}$ $= 10.76923 \text{ J } ^\circ\text{C}^{-1} \approx 10.8 \text{ J } ^\circ\text{C}^{-1}$	1M 1A 2	
(c) Since energy transferred by the boiling water is taken to be energy from the metal sphere / <u>extra energy is transferred (by the boiling water) to the cup of water /</u> the final temperature is higher. <u>The true value of C is smaller (than the calculated value).</u>	1A 1A 2	
(d) Not justified (with correct explanation). Copper – (good) conductor while polystyrene – (good) insulator or poor conductor, <u>more energy will be lost to the surroundings by conduction via a copper cup / most part of the polystyrene cup is still at room temperature, negligible energy is absorbed.</u>	1A 1A 2	

1. (a) conduction / radiation	1A 1	
(b) Will not, as the silvery surface of aluminum foil is a poor radiation absorber / good radiation reflector.	1A+1A 2	
(c) (i) The water droplets come from the moisture / water vapour in the air (surroundings).	1A 1	
(ii) $E = (0.40 \times 10^{-3})(2.26 \times 10^6)$ $= 904 \text{ J}$	1M 1A 2	
2. (a) (i) $E_K = \frac{1}{2} (6.63 \times 10^{-26}) (500)^2$ $= 8.2875 \times 10^{-21} \text{ J} \approx 8.29 \times 10^{-21} \text{ J}$	1M 1A 2	Accept: $(8.29 \sim 8.30) \times 10^{-21} \text{ J}$
(ii) $E_K = \frac{3RT_0}{2N_A}$ $= \frac{3(8.314)(298)}{2(6.022 \times 10^{23})}$		